# Loan Default Prediction

An interpretable credit-scoring model for home equity loan approvals

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

Shubham Garg | November 2026

# Key takeaways

**1 in 5**

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

**62%**

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is recorded. Unverified income is the strongest warning sign.

**93%**

of future defaulters are caught by the proposed Random Forest model, while 2 of every 3 applicants it flags really would have defaulted.

**~$9M**

indicative annual benefit on 6,000 applications, from roughly $10.8M of default-related cost down to about $1.8M.

# The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit; bad loans eat it up.
- Today each application is judged manually: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act every rejection needs a specific, defensible reason, so the model must be interpretable.

*Objective: predict which applicants are likely to default and tell the bank which attributes to weigh when approving a loan.*

**The data: Home Equity dataset (HMEQ)**

**5,960**
recent loans

**12**
attributes known at application

**19.9%**
defaulted or seriously delinquent

**44%**
of applications have a blank field

# Approach: from raw applications to a decision the bank can defend

1

## Explore

Univariate, bivariate and correlation analysis; default rate by band; pattern of missing values

2

## Prepare

Flag every missing field, impute median/mode, cap outliers for the linear model only, one-hot encode, 70/30 stratified split

3

## Model

Logistic Regression (linear benchmark), Decision Tree, Bagging, Random Forest; class weights; cross-validated grid search

4

## Measure

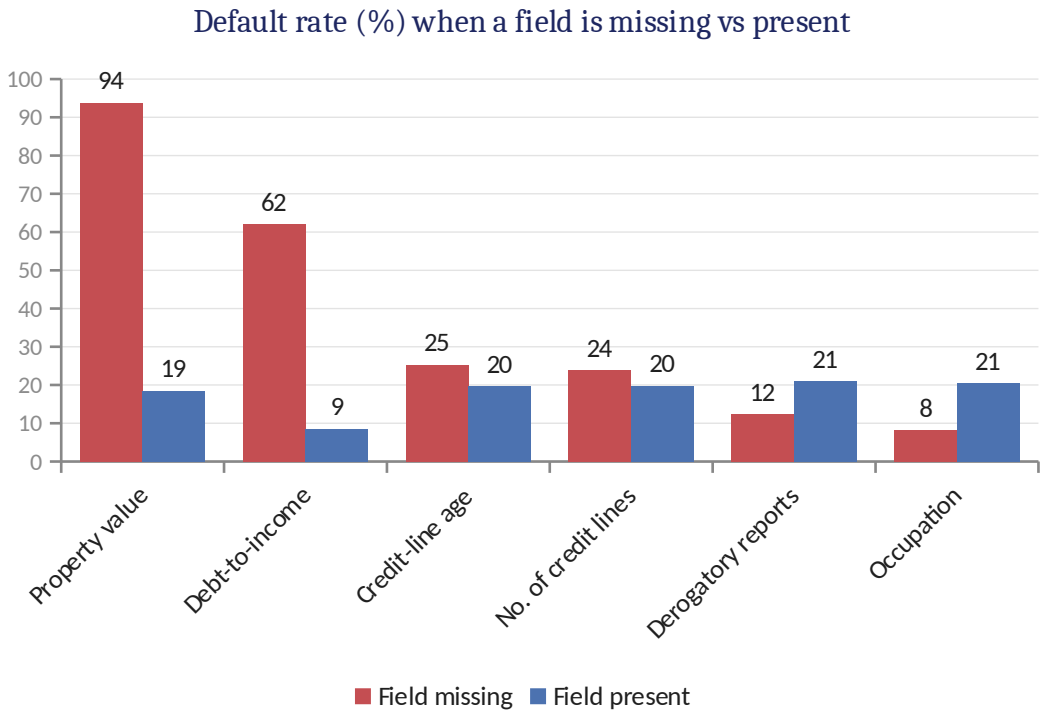
Recall on defaulters as the primary metric; precision, F1, ROC-AUC as guard-rails; cost-based threshold

5

## Recommend

Model specification, business rules, implementation plan, benefits, risks

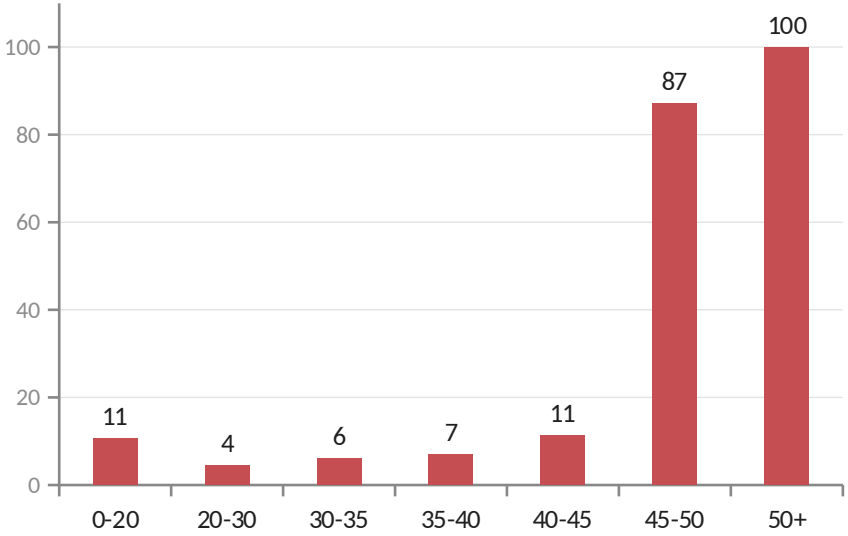
# Insight 1: information the applicant did not provide is the strongest warning sign



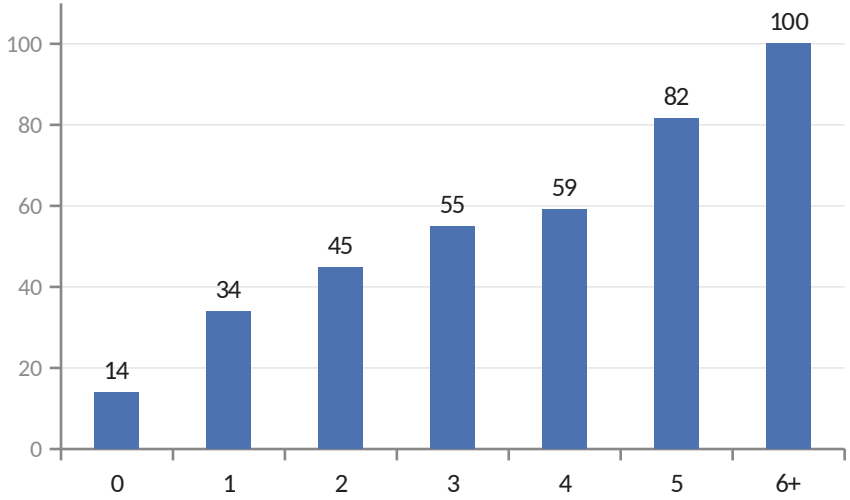

- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data).
- No property valuation: 94% default (112 applicants).
- Overall default rate is 20%, so these gaps multiply risk three to five times.
- Interpretation: income or collateral that could not be verified, or that the applicant chose not to supply.
- Action: every imputed field gets a 'was missing' flag in the model, and verification should become mandatory in the process.

# Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)



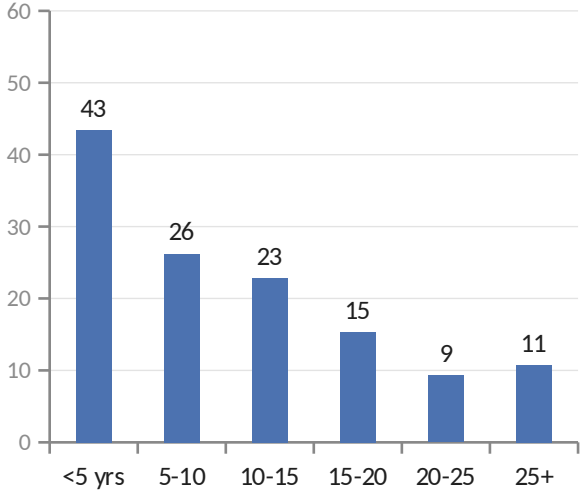

Default rate (%) by number of delinquent credit lines



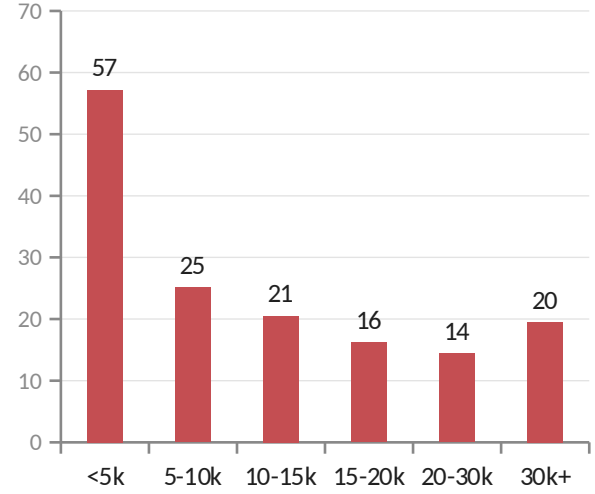

Below a 40% ratio only 4-7% default; above 45% almost everyone does. Each delinquent line or derogatory report roughly doubles the risk (0 reports: 17%, 1: 39%, 2: 51%, 3+: 70%+). Both are plain-language rules an applicant can be given.

# Insight 3: long credit histories protect, small loans do not, occupation matters less

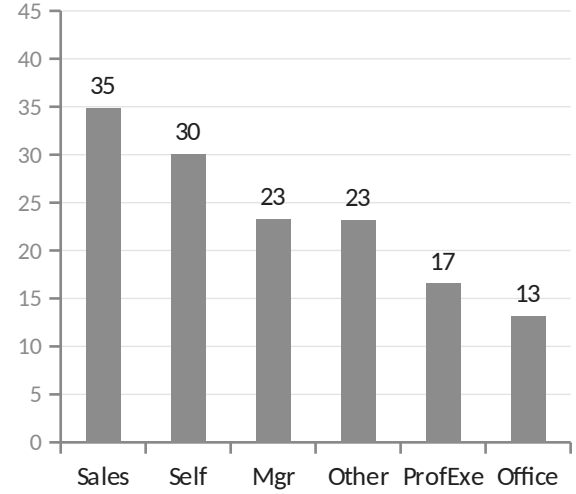
By age of oldest credit line




By loan amount ($)




By occupation




Default rate (%) by group. Applicants with under five years of credit history default four times as often as those with twenty-plus years. Loans under $5,000 default 57% of the time. Occupation shifts risk moderately but adds almost nothing once financial behaviour is known, so the final model does not need it.

# Measure of success: the two mistakes do not cost the same

### Missed defaulter

The model says 'will repay', the customer defaults.

**Bank loses most of the principal plus collection cost: about $9,000 on a typical $18,600 loan.**

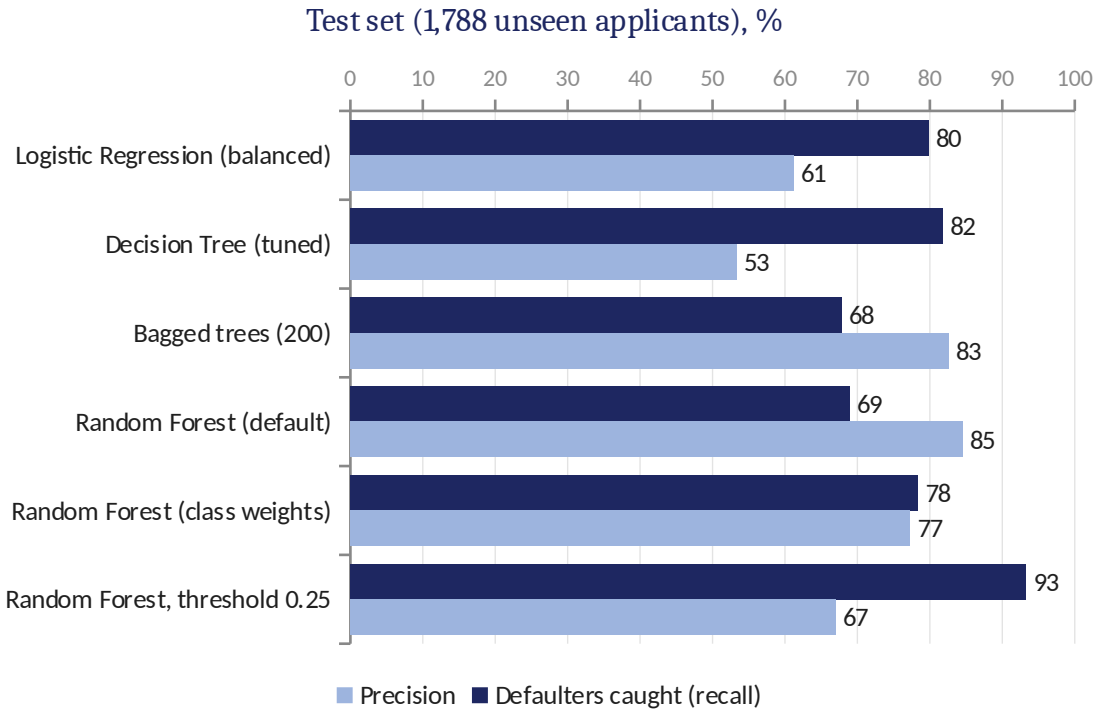
### Wrongly declined

The model says 'will default', the customer would have repaid.

**Bank forgoes the interest margin on one good customer: about $2,000.**

- A missed defaulter costs roughly 4.5 times a wrongly declined customer, so the primary metric is Recall: the share of future defaulters the model catches.
- Guard-rails: Precision (how many of the flagged applicants really default) and F1, so the model does not simply decline everyone. ROC-AUC measures how well applicants are ranked by risk, which matters when the threshold moves.
- All models are tuned by cross-validated grid search on recall and judged on a held-out 30% of applicants the models never saw.

# Model comparison: the Random Forest gives the best balance




- Linear vs non-linear: logistic regression reaches 80% recall but only 61% precision. The data is made of thresholds, which trees capture and a straight line cannot.
- Single tree vs ensembles: the tuned tree catches 82% but declines almost one good customer per defaulter caught (53% precision). Bagging 200 trees lifts precision to 83% and AUC to 0.95 at 68% recall.
- The class-weighted forest catches 78% at 77% precision, with the best ROC-AUC (0.96): it ranks applicants by risk most accurately.
- Because it ranks so well, lowering its threshold to 0.25 lifts recall to 93% at 67% precision, the lowest expected cost of any option.

# Proposed solution: a Random Forest scorer with a Decision Tree explanation layer

## Model specification

- Scoring engine: Random Forest, 200 trees, class weights 0.2 (repaid) / 0.8 (default).
- Inputs: the 10 financial attributes plus 'debt-to-income missing' and 'property value missing' flags. Occupation and loan purpose are not needed.
- Decision threshold 0.25 (cost-minimising) instead of the default 0.5; approve / refer / decline bands keep an underwriter in the loop near the threshold.
- Explanation layer: a pruned Decision Tree built on the same data. Its rules (ratio missing, ratio above 44%, delinquencies, derogatory reports, credit-history length) supply the adverse-action reasons the ECOA requires.
- Validated on a 30% hold-out; hyper-parameters chosen by stratified cross-validation.

**93%**
of defaulters caught

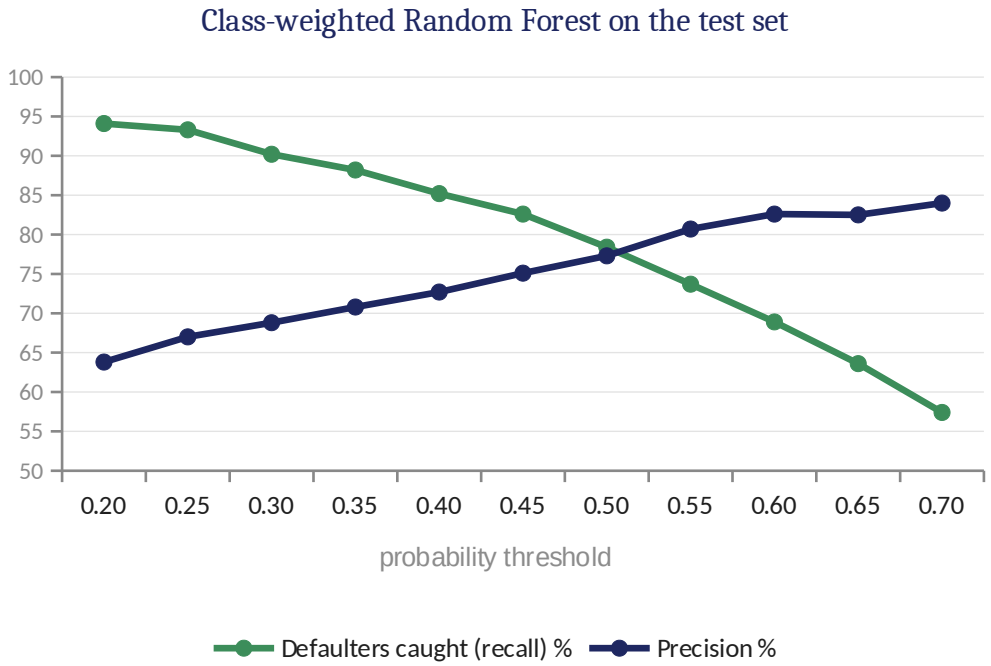
**67%**
of flagged applicants truly default

**0.96**
ROC-AUC: risk ranking quality

**90%**
overall accuracy

*Per 1,000 applicants: about 187 of 200 defaulters flagged in advance, 13 missed, 92 good customers referred or declined.*

# Why a 0.25 threshold: the decision cut-off is a business dial, not a constant




### At 0.50 (default)

- 43 missed defaulters
- 46 good customers declined
- $479k indicative cost

### At 0.25 (proposed)

- 13 missed defaulters
- 92 good customers declined
- $304k indicative cost

Per 1,000 applicants; cost assumes $9,000 per missed defaulter and $2,000 per wrongly declined customer. Thresholds 0.20-0.30 are all within 12% of the minimum, so the choice is robust.

# Business impact: an indicative $9 million a year

| Per year, 6,000 applications | Current process | With the proposed model |
|---|---|---|
| Loans that default | about 1,200 (20%) | about 80 slip through |
| Default-related losses | about $10.8M | about $0.7M |
| Good customers declined or referred | none | about 550, margin forgone about $1.1M |
| **Total cost of credit errors** | **about $10.8M** | **about $1.8M** |

**~$9M**

indicative gross annual benefit; about $8.5M after running costs

- Assumptions: 6,000 applications a year, average loan $18,600, loss given default about 50%, forgone margin about $2,000 per declined good customer, production performance equal to test performance.
- Costs: roughly $300-400k one-off (pipeline, integration, validation, fairness review, training) and about $100k a year to run and monitor. Pays back within the first year even if only a quarter of the benefit is realised.
- Non-financial: faster decisions, consistent treatment, auditable reasons for every decline, verified income and valuation on every application.

# Recommendations and implementation roadmap

### 1 Months 0-2: prepare

- Make income and valuation verification mandatory fields
- Build the scoring pipeline and reason codes
- Calibrate the threshold with real loss and margin figures
- Fairness and model-risk review

### 2 Months 2-4: shadow and pilot

- Score every application alongside the manual decision
- Compare outcomes, tune the approve / refer / decline bands
- Train underwriters on reason codes and referrals

### 3 Month 4 onwards: scale

- Switch on automated approve / decline for clear cases
- Monthly monitoring of recall, precision, approval rate, missing fields
- Yearly re-training; threshold review with finance

## Policy rules to adopt now, independent of the model

- Decline or escalate when the debt-to-income ratio is above 45%, when there are two or more delinquent credit lines, or two or more major derogatory reports.
- Treat an application with no verifiable income or property valuation as high risk, never as neutral.

# Risks, challenges and further analysis

## Risks and mitigations

- Selection bias: data covers approved loans only. Mitigate with a shadow period on all applications and reject inference.
- The 'ratio missing' signal will change once verification is mandatory. Re-train after the rule is in place.
- Economic drift: monitor monthly, re-train yearly, review the threshold.
- Fair lending: keep occupation out of the model; disparate-impact testing before launch and annually.
- Explainability and automation bias: tree-based reason codes, a referral band, documented overrides.

## Further analysis

- Reject inference to cover applicants the old process declined.
- Engineered features: loan-to-value and payment-to-income ratios.
- Gradient boosting and probability calibration, so a score of 0.25 means a 25% chance of default.
- Time-based validation once application dates are available.
- Risk-based pricing for the referral band; early-warning scoring on existing loans.
- Per-applicant SHAP-style reason codes in applicant-friendly language.

# Thank you

Questions and discussion

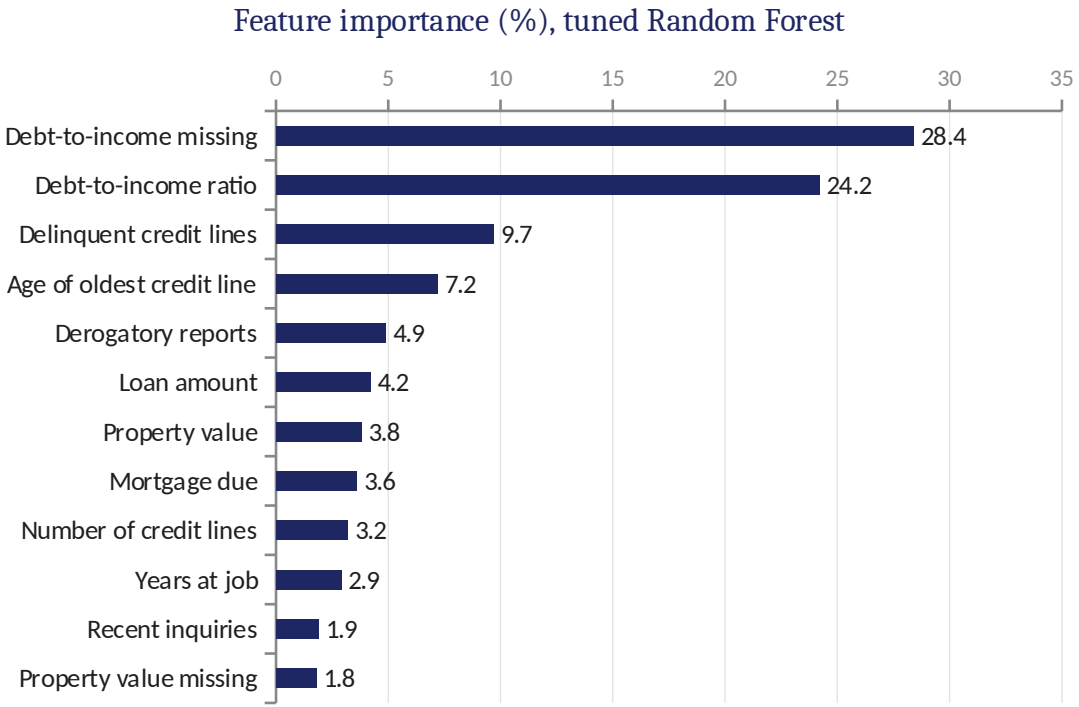
Catch 93% of defaulters, explain every decision, and save an indicative $9M a year.

# Appendix A: all models on the held-out test set

| Model | Recall | Precision | F1 | Accuracy | ROC-AUC | Train recall |
|---|---|---|---|---|---|---|
| Logistic Regression (plain) | 0.569 | 0.772 | 0.655 | 0.880 | 0.907 | 0.643 |
| Logistic Regression (balanced) | 0.798 | 0.612 | 0.693 | 0.859 | 0.906 | 0.823 |
| Decision Tree (default) | 0.686 | 0.706 | 0.696 | 0.880 | 0.807 | 1.000 |
| Decision Tree (class weights) | 0.616 | 0.728 | 0.668 | 0.878 | 0.779 | 1.000 |
| Decision Tree (tuned: balanced, min 20 per leaf) | 0.818 | 0.533 | 0.645 | 0.820 | 0.879 | 0.924 |
| Bagging (200 fully grown trees) | 0.678 | 0.826 | 0.745 | 0.907 | 0.946 | 1.000 |
| Random Forest (default, 200 trees) | 0.689 | 0.845 | 0.759 | 0.913 | 0.962 | 1.000 |
| Random Forest (class weights 0.2/0.8) | 0.784 | 0.773 | 0.779 | 0.911 | 0.958 | 1.000 |
| Random Forest (tuned for recall) | 0.784 | 0.653 | 0.712 | 0.874 | 0.933 | 0.921 |
| Random Forest (tuned, 12 features) | 0.782 | 0.664 | 0.718 | 0.878 | 0.928 | 0.917 |
| **Proposed: RF class weights, threshold 0.25** | **0.933** | **0.670** | **0.780** | **0.895** | **0.958** | - |

Train recall of 1.000 marks the un-pruned models that memorised the training data. Grid search used stratified cross-validation on the training set only, scoring on recall.

# Appendix B: what the Random Forest relies on




- The debt-to-income ratio and its missing flag carry more than half (53%) of the forest's decision power.
- Next come repayment history (delinquencies, derogatory reports) and depth of credit history.
- Loan, property and mortgage amounts matter in combination but are weak on their own.
- Occupation and loan-purpose dummies rank at the bottom; removing them and nine minor flags left performance unchanged (12-feature model).
- Logistic-regression coefficients and the decision-tree rules point to the same drivers, so the signal is stable across techniques.

# Appendix C: the explanation layer, rules read from the pruned Decision Tree

| | | |
|---|---|---|
| 1 | **Is the debt-to-income ratio missing?** | Yes: highest-risk branch (62% historical default). Verify income or decline. |
| 2 | **If recorded, is the ratio above about 44%?** | Yes: almost certain default in the data (87-100%). Decline. |
| 3 | **Any delinquent credit lines?** | One or more pushes the applicant back into the risky branch, even with a healthy ratio. |
| 4 | **Any major derogatory reports, and how long is the credit history?** | Short history (under about 15 years) with a derogatory report is risky; long clean histories are safe. |
| 5 | **Loan size, mortgage balance and property value** | Refine the decision deeper in the tree; small loans with weak history lean towards default. |

Tuned tree: Gini criterion, balanced class weights, minimum 20 applicants per leaf, depth 14, 105 leaves. Test recall 82%, precision 53%: used for explanations, not for scoring.

# Appendix D: data treatment and pre-processing

| Step | What was done | Why |
|---|---|---|
| Duplicates | None found; no rows dropped | 5,960 loans retained, including 2 with only the loan amount recorded |
| Missing values | A 0/1 'was missing' flag per column, then median (numeric) or mode (categorical) imputation | Missingness of debt-to-income and property value is the strongest default signal; dropping rows would lose 44% of the data |
| Outliers | IQR capping for the linear model only; DEROG and DELINQ capped at the 99th percentile; tree models use raw values | Linear models are pulled by extremes (debt-to-income 203%, credit history 97 years); trees split on rank order and keep the 'above 50% always defaults' signal |
| Encoding and scaling | One-hot encoding of REASON and JOB (drop first); standard scaling for the linear model only | Comparable coefficients and stable convergence for logistic regression |
| Split | 70% train / 30% test, stratified on the target, fixed seed; identical applicants in both data versions | Both sets keep the 20% default rate; results comparable across models |
| Imbalance | Class weights (balanced or 0.2/0.8) inside the models rather than resampling | Keeps every real applicant; lifted recall by 23 points for logistic regression and 10 for the forest |